

Organic Farm Pest Management AI System

1. Project Scenario

Organic farming represents a growing sector of agriculture, with farmers committed to sustainable practices that avoid synthetic pesticides and fertilizers. However, pest management remains one of the most challenging aspects of organic farming, requiring:

- Quick and accurate pest identification to prevent crop damage
- Knowledge of organic-approved treatment methods
- Understanding of integrated pest management (IPM) principles
- Timely intervention to minimize economic losses

2. Problem Statement

Many organic farmers, especially small-scale and emerging farmers, face significant challenges:

Knowledge Gap: Limited expertise in pest identification leads to delayed or incorrect treatments, resulting in crop losses averaging 15-25% annually.

Remote Locations: Many farms operate in areas with poor or no internet connectivity, making online resources inaccessible during critical decision-making moments.

Time Sensitivity: Pest outbreaks can spread rapidly; farmers need immediate identification and treatment guidance, often outside business hours when agricultural extension services are unavailable.

Resource Constraints: Small organic farms cannot afford expensive pest management consultants or laboratory identification services.

Certification Requirements: Organic farmers must use only approved treatment methods to maintain their certification, requiring specialized knowledge of organic pest control options.

3. Project requirements

- You will develop an intelligent, conversational AI system with advanced computer vision capabilities that serves as a comprehensive pest management consultant for organic farmers. The system will operate entirely offline on edge computing devices, enabling farmers to capture or upload images of crop pests and receive immediate, expert-level identification along with tailored organic treatment recommendations.
- You need to use your well-developed or fine-tuned models during this course in this project.